

Ecosystem functions and functional traits for the study of phytophagous scarab beetles (Coleoptera: Scarabaeidae)

Funciones ecosistémicas y rasgos funcionales para el estudio de los escarabajos fitófagos (Coleoptera: Scarabaeidae)

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Abstract

1. Owing to the rapid and ongoing anthropogenic changes impacting ecosystem functions and their interplay with biodiversity, there is a growing focus on assessing insect functional diversity. This emphasis helps to unravel key roles of insects on ecosystem processes and to identify the assembly mechanisms that regulate species coexistence.
2. The ecological functions of phytophagous scarab beetles (Scarabaeidae) have been relatively unexplored, and potential functional traits are scattered throughout the literature. Species within this group exhibit diverse lifestyles, forms and sizes. They exploit various resources such as decomposing wood, foliage, roots and flowers, with a small subset identified as predators. Although many of their representatives are significant agricultural pests, information on their contributions in natural ecosystems has not been consolidated.
3. This study reviews the ecological functions potentially carried out by phytophagous scarab beetles and proposes functional traits to be incorporated into studies of functional diversity. Through this, the goal is to integrate the scattered information, stimulate the generation of questions that go beyond classical ecological theory and contribute to advancing our knowledge of their ecosystem functions, as well as the foundations for understanding community structure. We propose to use trophic groups to define functions such as bioturbation, organic matter decomposition, biological control and potential pollination.
4. A simplified list of functional traits was obtained by conducting correlation analyses using a trait matrix from a region in northern Colombia. This represents a significant advance for the study of traits within the framework of functional ecology, applicable to territorial planning and biodiversity conservation.

KEYWORDS

'Pleurosticti' scarab beetles, biological control, bioturbation, functional diversity, organic matter decomposition, pollination

INTRODUCTION

Land use changes have a significant impact on the diversity of terrestrial insects, leading to species extinction and population declines (Cardoso et al., 2020; Córdoba-Aguilar & Roitberg, 2021; Sánchez-Bayo & Wyckhuys, 2019; van der Sluijs, 2020). To assess these impacts, studies on biodiversity have introduced an additional component known as functional diversity, which allows us to determine changes in ecosystem properties and consequently improve our understanding of ecosystem functions and services (Casanoves et al., 2011; de Bello et al., 2021). Functional diversity is the variety of functional traits among the species that coexist in ecological communities, supposedly reflecting the diversity of functional roles in ecosystem processes (Moreno et al., 2018). This dimension of biodiversity involves the measurement of traits reflecting the organism's performance in a given environment (Tilman, 2001). Functional traits are measurable biological characteristics (morphological, phenological, physiological or behavioural) that influence the performance of organisms and can be related to ecosystem processes (matter cycling and energy flow), ecosystem stability (resistance and resilience), biological interactions (intra- and interspecific) or habitat modification (Dawson et al., 2021; Gonçalves-Souza et al., 2023; Violle et al., 2007). Traits are categorised into response traits, which have an impact on organism's fitness and are affected by environmental factors, and effect traits, which have an impact on ecosystem functions (Lavorel & Garnier, 2002; Violle et al., 2007), although in many cases, the same trait can be both a response and an effect trait (Gonçalves-Souza et al., 2023). The selection of a suitable set of functional traits contributes to a more accurate description of ecosystem functionality, and this selection is considered the cornerstone of functional diversity studies (Dawson et al., 2021; de Bello et al., 2021; Gonçalves-Souza et al., 2023).

Understanding the effects of habitat transformations on the functional diversity and functional traits of insects is essential to guide conservation efforts towards an approach that prioritises ecosystem functions and stability, moving beyond species lists and numerical counts (Cadotte et al., 2011). In landscapes characterised by habitat loss, fragmentation and limited connectivity, species are not equally affected. In fact, degraded landscapes promote the invasion and expansion of species directly affected by the frequency of the relevant effect traits, whereas other species with response functional traits, that make them susceptible to disturbances, tend to be lost (Tschamtké et al., 2012). Therefore, in recent years, the study of functional traits has been incorporated into diversity assessments to address questions related to the consequences of environmental transformation on the functions performed by species in ecosystems.

Research linking the traits and functional performance of terrestrial beetles has primarily focused on Carabidae and coprophagous Scarabaeidae families. Carabid beetles are mainly predators and are commonly used as a subject of study for the analysis of functional diversity, especially in Europe and Asia (Baulechner et al., 2020; Feng et al., 2021; Gobbi et al., 2015; Liu et al., 2014; Micó et al., 2020; Nolte et al., 2017; Pakeman & Stockan, 2014; Sipos et al., 2017).

Coprophagous beetles mainly feed on dung and carrion and represent the coleopteran group that has undergone the most extensive trait and functional diversity studies in the Neotropics (Barragán et al., 2011; Beiroz et al., 2018; Ortega-Martínez et al., 2020; Rivera et al., 2021). For both carabids and dung beetles, studies have focused on describing functional diversity patterns in landscapes. However, the ecological functions of other members of the Scarabaeoidea superfamily, such as phytophagous Scarabaeidae, have been relatively unexplored, and the traits that could influence their performance in the ecosystem have not yet been thoroughly investigated and remain scattered in the literature.

According to the classification of Bouchard et al. (2011), within the family Scarabaeidae, we found phytophagous beetles belonging to the subfamilies Cetoniinae, Rutelinae, Dynastinae and Melolonthinae, which exhibit a wide range of colours, sizes, habits and lifestyles. Even when they are commonly known as phytophagous scarab beetles, most adult species are highly associated with the consumption of plant material, ingesting different parts of the plant such as leaves, plant secretions, fruits, stems or flowers. Only a few species may prey on other insects. Larval diets can vary, including roots, decomposing plant tissues, leaf litter and humus (Morón et al., 1997). Because of this polyphagy, their tremendous diversity cannot be solely explained by insect–host plant co-diversification, which is a widely accepted hypothesis for the great species diversity in phytophagous insects (Eberle et al., 2014). Many of their representatives are recognised as significant agricultural pests given that habitat homogenisation, resulting from agriculture, has increased their abundance and biomass (García-Atencia & Amat-García, 2021).

In the late 1990s and the early 2000s, a significant number of publications emerged to understand various aspects of the biology and ecology of phytophagous scarab beetles. The objective was to generate information that would enable the identification, quantification and prevention of agricultural damage. The initial contributions from the Neotropics included descriptions of the life cycle of *Phyllophaga* Harris (Coleoptera: Scarabaeidae) species associated with maize cultivation in Mexico (Marín Jarillo, 2001; Ramírez-Salinas & Castro-Ramírez, 2000; Villalobos, 1999) as well as faunal lists from different regions of the same country (Aragón et al., 2001; Morón-Ríos & Morón, 2001; Pardo et al., 2005; Villegas et al., 2008). Additionally, there have been data on the reproductive behaviour of the species of interest (Oliveira & García, 2003). Moreover, Deloya et al. (2007) proposed functional groups and guilds of Scarabaeidae species, based on their feeding habits during their larval and adult stages. This reveals the complexity of resources used by these beetles and suggests important processes that are often hidden and may be affected by biodiversity loss. However, dietary habits do not necessarily reveal morphological differentiation within groups; in some lineages, resource partitioning allows species coexistence, and this selective pressure may have led to morpho-spatial divergence in closely related groups (Eberle et al., 2014). Consequently, segregating communities solely by trophic groups hinders our understanding of how the functional niche is distributed and how community assembly occurs. Therefore, it is crucial to compile information about the biology,

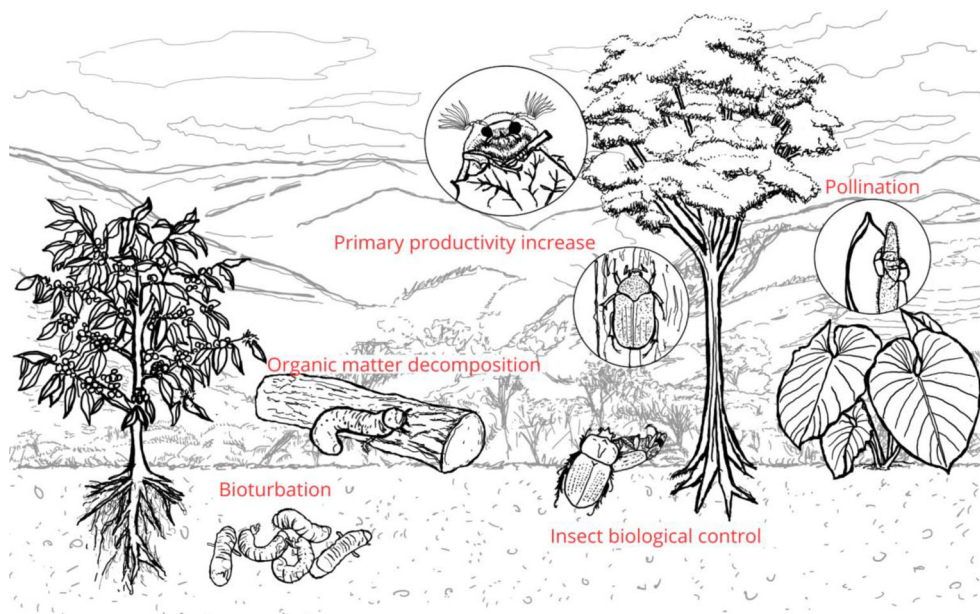


FIGURE 1 Ecosystem functions carried out by phytophagous scarab beetles (Scarabaeidae). Illustration by Anibal Maldonado Currea.

evolution and natural history of phytophagous scarab beetles to elucidate their role in ecosystem functions and identify traits such as varying body structure sizes, which can be response traits affected by environmental changes or effect traits having an impact on ecosystem processes.

This study aimed to review the ecological functions of phytophagous scarab beetles and propose functional traits to be incorporated in studies of their functional diversity. The goal was to integrate scattered information about this group of beetles, encourage the generation of questions that go beyond the classical ecological theory and advance our knowledge on the ecosystem functions provided by this group to promote biodiversity conservation. Once the functions were identified, a broad list of functional traits was proposed along with their respective justifications. In addition, a list was proposed to facilitate the analysis of functional diversity using a trait matrix for 32 species of the four subfamilies from northern Colombia. From this data matrix, Spearman's correlation tests were conducted to eliminate highly correlated traits and thus avoid collinearity. In addition to the statistical exclusion criteria, functional justification of the selected trait pool was considered. Therefore, this study provides a trait-based framework of phytophagous scarab beetle traits to promote their use beyond traditional species-based studies.

ECOSYSTEM FUNCTIONS ASSOCIATED WITH PHYTOPHAGOUS SCARAB BEETLES

The contribution of phytophagous scarab beetles to ecosystem functions can be inferred from their feeding habits and substrate usage for oviposition. For example, nectar consumption can be associated with pollination and by processing wood for larval development, some

species are linked to organic matter decomposition. Therefore, based on the compilation of their natural history, we list ecosystem functions potentially associated with phytophagous scarab beetles, including bioturbation, organic matter decomposition, pollination, increased primary productivity and insect biological control (Figure 1). Some of these ecosystem functions directly or indirectly contribute to human well-being and quality of life, so they can be recognised as ecosystem services or nature's contributions to people (Díaz et al., 2018). As far as we know, the outlined relationships between phytophagous scarab beetles and the ecosystem functions described below have not been experimentally tested, so they represent knowledge gaps that challenge further research.

Bioturbation

Bioturbation is defined as the displacement and mixing of sediment particles by animals and plants, which can influence the soil biota and productivity by increasing aeration and porosity (Nichols et al., 2008). In natural ecosystems, where resource heterogeneity allows for natural population control, larval development of beetles in the soil generates bioturbation processes that contribute to improving the physical and chemical properties of these soils.

In all subfamilies of phytophagous Scarabaeidae, there are species that feed on roots (rhizophagous) during their larval stage. Owing to their impact on economically important plants, they have historically been associated as pests of agricultural crops. In the case of Melolonthinae beetles, larvae feed on the tender and bulbous roots of various plants, leading to significant reductions in crop yields. Within this group, genera of dynastine beetles such as *Eutheola* Bates, *Ligyris* Burmeister, *Cyclocephala* Dejean, *Dyscinetus* Harold and *Stenocrates*

Burmeister and ruteline beetles of the Anomalini tribe have been reported as root-eating beetles (Pardo-locarno et al., 2012).

According to Tapia-Rojas et al. (2013), in cultivated and pristine soils, beetles feeding on roots are beneficial as they help improve soil structure by favouring water and airflow. In doing so, they incorporate their excreta, which is rich in assimilable nitrogen and abundant micro-organisms that benefit plants. Therefore, Romero-López et al. (2010) argued that the work performed by these species contributes to (i) increasing soil porosity, drainage and aeration by behaving similarly to earthworms and positively influencing the physical characteristics of the soil; (ii) improving soil chemical properties by immobilising and distributing nutrients, as well as mineralizing and/or humifying organic matter, thus functioning as biological catalysts and (iii) stimulating various soil biological activities, enhancing the microbiological decomposition of organic matter, regulating biotic composition and exporting some soil nutrients in their adult phase to the upper soil subsystem. Considering the aforementioned, identifying meaningful traits of root-eating scarab beetles and assessing their role in soils will allow us to understand and quantify their contribution of bioturbation in landscapes with land use changes.

Organic matter decomposition

Decomposition of organic matter occurs in ecosystems through organisms feeding on decaying organic matter (saprophagous species). This process is carried out by different species that generate temporal succession in resource partitioning based on the degree of decomposition (Stokland et al., 2012). As resources advance in their stages of degradation, the soil retains more moisture and some soils become suitable temporary microhabitats. Therefore, many saprophagous insects are highly specialised in specific resources or stages of decomposition (Stokland et al., 2012).

Phytophagous scarab beetles mainly participate in this process during their larval stage, thereby contributing not only to organic matter decomposition but also to nutrient cycling (Morón et al., 1997). This has a positive impact on soil fertility, as high levels of decomposed organic matter increase carbon reserves, creating optimal conditions for seed establishment (FAO, 2017).

Some Dynastinae larvae can break down large amounts of leaf litter, transforming it into organic matter that contributes to soil fertility (Morón, 1985). However, some species may have facultative habits that depend on the amount of organic matter present in the soil. Lugo-García et al. (2017) argued that the scarcity of organic matter and the abundance of roots in agricultural plots promote the predominance of root-eating larvae, whereas in soils rich in organic matter and high root density, root-eater and saprophagous larvae coexist. For example, *Cyclocephala lunulata* Burmeister was reported to feed on roots in corn crops in Mexico (Lugo-García et al., 2017), but at the same time, it exhibited saprophagous behaviour on sugarcane crops in the same country (Deloya, 1998). The feeding habits of the species *Lygirus salli* Bates was also associated with agricultural residues and dung rather than roots, where it is typically found (Lugo-García et al., 2017).

Cetoniines and rutelines have saprophagous members in their larval stage that exploit resources, such as soil organic matter, leaf litter or decomposing logs (Deloya et al., 1995; Morón & Deloya, 2002; Neita-Moreno et al., 2006). Although adult Cetoniinae have specialised jaws for consuming soft foods, such as nectar, sap, pollen and ripe fruit, the larvae are characterised by strong jaws that aid in the decomposition of plant-derived organic matter (Deloya & Gasca-Álvarez, 2018). Larvae of beetles from Rutelinae, Dynastinae and Cetoniinae have been found in association with ant refuse dumps of the genera *Atta* Fabricius (Hymenoptera: Formicidae) and *Acromyrmex* Mayr (Hymenoptera: Formicidae) (Navarrete-Heredia, 2001), and larvae of Cetoniinae have been reported in nests of *Cornitermes* Wasmann (Isoptera: Termitidae) and *Dwarsitermes* (Silvestri) (Isoptera: Termitidae) termites (Puker et al., 2012), likely taking advantage of the microclimatic conditions and food (mostly faeces) generated in these spaces beneath the soil surface. In an experiment conducted by Micó et al. (2011), the larvae of a *Cetonia* species were able to degrade wood in a way that produced nutrient-rich residues with easily decomposable lignin. This facilitates the decomposition of detritus by other saprophagous organisms, thus contributing to the carbon cycling system. In contrast, rutelines are commonly found on fallen logs, feeding on partially decomposed wood or accumulated detritus (Morón et al., 1997). However, in this group, identification from immature stages can be complex, so the recognition of their contribution to this function in the larval stage is still limited.

Pollination

The role of phytophagous scarab beetles in pollination is poorly studied. Some adult beetles of the Dynastinae subfamily feed on nectar or pollen and are frequently associated with inflorescences (Moore & Jameson, 2013), suggesting their potential role as pollinators. They have been categorised as meliphagous (Deloya et al., 2007). Associations with the inflorescences of Araceae have been observed in the Cyclocephalini tribe. These inflorescences, with their spathe and spadix, may provide attractive conditions for these species by offering stable sources of food and reproduction areas (García-Robledo et al., 2004; Gasca-Álvarez, 2013; Maia et al., 2012; Milet-Pinheiro et al., 2017; Moore & Jameson, 2013).

Neotropical species of Cyclocephalini contribute to the pollination of early diverging angiosperms, such as Nymphaeaceae, Annonaceae, Magnoliaceae, Araceae, Cyclanthaceae and Arecaceae (Moore & Jameson, 2013). Species of the genus *Cyclocephala* are common visitors to radially symmetrical flowers and consume the fruits of a wide variety of plants. It has been reported that the reproduction of species in this genus is linked to the flowering period, and their presence in Annonaceae flowers can significantly contribute to pollination (Costa et al., 2017; Gottsberger & Webber, 2018; Saravy et al., 2022).

The diet of many adults in subfamily Cetoniinae consists of ripe or decomposing fruits, pollen, nectar and floral parts from various plant species, and some species consume sugary liquids secreted through wounds or breaks in branches and trunks (Di Iorio, 2014;

Suárez & Amat-García, 2007). Generally, they are associated with the trophic group of frugivores. Despite the importance of some species in this subfamily as potential pollinators, their functional roles in pollination are not well known. Studies conducted in Japan found that the species *Protaetia cataphracta* Arrow and *Eucetonia pilifera* Motschulsky transport a large amount of cross-pollinated pollen with high genetic diversity of Magnoliaceae species. This family is characterised by a reproductive system highly specialised in cooperating with beetles, so it is argued that, in return, they contribute significantly to the reproduction of the plant (Matsuki et al., 2008).

In addition, Cetoniinae beetles have been closely associated with the reproduction of Orchidaceae plants, whose floral scent and spur length appear to be important traits for floral visits (Peter & Johnson, 2009, 2014). In fact, there is discussion about the shift from bee pollination to beetle pollination as a driver of floral divergence in a species complex within this plant family (Peter & Johnson, 2009, 2014; Sugiura et al., 2021).

Primary productivity increase

Herbivory is the most common feeding habit among adult species of the four subfamilies of Scarabaeidae outlined in this study, which represent strict phytophagous. However, as a result of phytophagy, stem-feeding species are found, which feed on tender plant stems. Although this trophic habit can have a negative impact in plants, in some cases, herbivory enhances plant fitness through overcompensation, that is, plants have higher fitness when they are damaged by herbivores, compared with related plants that are undamaged (Agrawal, 2000). Thus, herbivory can stimulate plant biomass production in natural ecosystems. This is achieved by removing older tissues with lower photosynthetic rates, which helps to reduce leaf-ageing rates, thereby prolonging active photosynthesis. This significantly contributes to the

survival and vigour of plant species (Granados-Sánchez et al., 2008). Although herbivory may increase plant biomass by removing senescent tissue or by reducing plant competition, the overcompensation theory has been debated (Belsky, 1986; Agrawal, 2000). Therefore, testing the potential mutualism between herbivore scarab beetles and plants remains as a research gap to be filled.

Insect biological control

One of the major contributions of predation in its broad sense is population control of other species via reducing the number of individuals or biomass. However, predation is limited to a few species within the genus *Phileurus*. For this group, there are records of consumption of other insects, such as *Heterogomphus chevrolatii* (Morón et al., 1997) and lepidopteran larvae (McCleve, 2007). Additionally, Ratcliffe et al. (2020) described beetles feeding on dung beetles and adult *Golofa eacus* beetles. However, it is unclear whether this behaviour extends to other tribes or is a derived and unique condition within Phileurini.

PROPOSAL OF FUNCTIONAL TRAITS FOR USE IN ECOLOGICAL STUDIES WITH PHYTOPHAGOUS SCARAB BEETLES

Based on the evolutionary analysis conducted by Eberle et al. (2014), the species traits considered to identify functional groups of Dynastinae in Panama (Basset et al., 2023) and the compilation of natural history data of phytophagous Scarabaeidae, this study proposes an extensive list of 16 morphological traits derived from morphological measurements (Figure 2), plus biomass and the trophic habits of adults (Table 1).

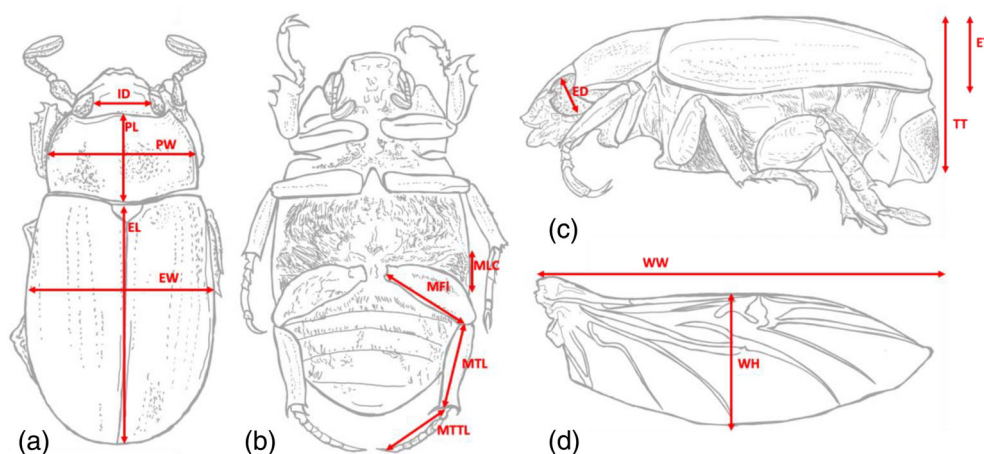


FIGURE 2 Proposed functional traits of phytophagous Scarabaeidae. Dorsal view (a), ventral view (b), lateral view (c) and wings (d). Morphological structures proposed as functional traits are also indicated. ED, eye diameter; EL, elytral length; ET, elytral thickness; EW, elytral width; ID, interocular distance; MFL, maximum metafemur length; MLC, maximum length of metacoxa; MTL, metatibia length; MTTL, metatarsus length; PL, pronotum length; PW, pronotum width; TT, total thickness; WH, wing height; WW, wing width.

TABLE 1 Proposed functional traits for phytophagous scarab beetles (Coleoptera: Scarabaeidae) and their functional justification.

Functional trait	Abbreviation	Attribute	Justification	
Body size	TL	mm	The body size has been used as a predictor of microhabitat, as well as a response trait to understand how the community responds to environmental changes. In addition, it has been utilised as a useful effect trait for various trophic groups (Micó et al., 2020).	
	TT	mm	Thickness may be related to the way the beetle perches or uses the resource.	
Pronotum	PL	mm	The pronotum has been used as an estimation of the body size of beetles (Creighton, 2005; Zhang et al., 2019).	
	PW	mm	Eye morphology can be influenced by environmental conditions, reflecting prey and/or predator recognition (Dias et al., 2017).	
Eyes	ID	mm		
	ED	mm		
Elytra	EL	mm	This trait is associated with the shape of the elytra, which is linked to the ability to fly and the type of flight of the insect. It also provides an idea of the shape of the posterior part of the body, its resistance during flight and the environments where a species may or may not be found, indirectly contributing to ecosystem functions in a given environment (Micó et al., 2020).	
	EW	mm		
	ET	mm		
Legs	LeBoRat	-	Thickness may be related to how the beetle positions itself to access resources (Eberle et al., 2014).	
			Leg length tends to be positively related to various movement-related characteristics and has an impact on energy expenditure. Individuals with legs longer than their body would have a greater ability to move quickly and efficiently (Arnold et al., 2017).	
Wings	MLC	mm	The presence of an expanded metacoxal process allows for a wider rotation of the hind limbs (Eberle et al., 2014).	
	WA	mm ²	It is directly associated with each species' dispersal capacity, determining their type of flight: cruising flight versus perching. Thus, species with a larger wing area use cruising flight, allowing them to make extensive movements, whereas smaller species use perching flight and tend to make more confined movements (Giménez, 2019).	
	WL	mg/mm ²		
	WiHeRat	-		
	Biomass	WW	mm	It is considered a key variable in community ecology, primarily in terms of energy flow, productivity and food web dynamics (Brown et al., 2004).
		WH	mm	
BIO		mg		
Adult guild	TrGr	1 = Phytophagous 2 = Meliphagous 3 = Stem-feeding 4 = Frugivores 5 = Predator	Trophic groups are associated with the use of resources by beetles in assemblages, and they can be used to analyse the impact of habitat transformation (Micó et al., 2020).	

Selection of a minimal pool of traits

As a case study to select a proper set of traits, a trait matrix was generated for 32 phytophagous scarab beetle species obtained from the entomological collection of the University of Atlántico UARC. The individuals are part of the material collected in various locations in the Serranía del Perijá, northern Colombia, by the first author (Figure 3). Ten individuals from each species were selected for traits measurements (Table S1). For those with low abundances, a minimum of three individuals were chosen. In species with high sexual dimorphism, half of the individuals selected were females and the other half were males.

With the traits described in Table 1, Spearman's correlation analyses were conducted to detect collinearity and thus generate a derived pool based on high correlations ($p < 0.05$). As a result, total length (TL), pronotum length (PL), pronotum width (PW), interocular distance (ID), elytral length (EL), maximum length of metacoxa (MLC), elytral width (EW), total thickness (TT), elytral thickness (ET), wing height (WH), wing width (WW) and wing area (WA), which exhibited high correlation values (Figure 4), were excluded. Eye diameter (ED), leg length to body length ratio (LeBoRat), biomass (BIO), wing loading (WL), wing width to height ratio (WiHeRat) and trophic group (TrGr) were retained because of their lower correlation values. Thus, for the study case used here, we propose these six traits as a minimal pool of traits for phytophagous scarab beetles in this region. In the following paragraphs, we discuss (1) their potential role in ecological functions as effect traits, (2) an example of its use as response traits, (3) their advantages and (4) some other characteristic not considered here that could be added as traits.

First, our minimal set of traits eliminates the bias of traits that contribute disproportionately to the same ecological functions. ED is

a response trait as eye morphology can be influenced by environmental conditions, but it is also an effect trait during the recognition and location of food and predators (Dias et al., 2017). The leg length to body length ratio is a trait directly related to locomotion, as beetles with legs longer than their body have a greater ability to move quickly and efficiently (Arnold et al., 2017). Biomass is a critical response trait related to the availability of resources in the environment, the pressure of infections and disease and climatic conditions, as it can be related to physiological responses such as beetle thermal tolerance and desiccation resistance (Nervo et al., 2021). Biomass is also an important effect trait as it has an influence at the ecosystem level, in terms of energy flow, productivity and food web dynamics (Brown et al., 2004). Wing loading and width-to-height ratio are related to beetle dispersal ability and flying mode. For example, species with a larger wing area can make extensive movements, whereas species with smaller wings tend to make more confined movements (Giménez, 2019). Finally, the presence and relative abundance of trophic groups depend on food availability and ecological interaction, so it can be used not only as a response trait to analyse the impact of habitat transformation (Micó et al., 2020) but also as an effect trait due to the potential relationship with the ecosystem functions described above (Figure 1).

Second, to exemplify how this set of traits can be used as response traits, we show their variation between two contrasting land covers: coffee crops and forest sites in our study case in the Serranía del Perijá, Colombia (Figure 5). The data were collected by the first author as part of her Doctoral thesis. Beetles were sampled in four crops and four forest sites using fruit and light traps between 2021 and 2022. The values of traits correspond to 14 and 16 species collected in forest and crops, respectively. This limited data set show some trends in species traits. However, their role as response traits

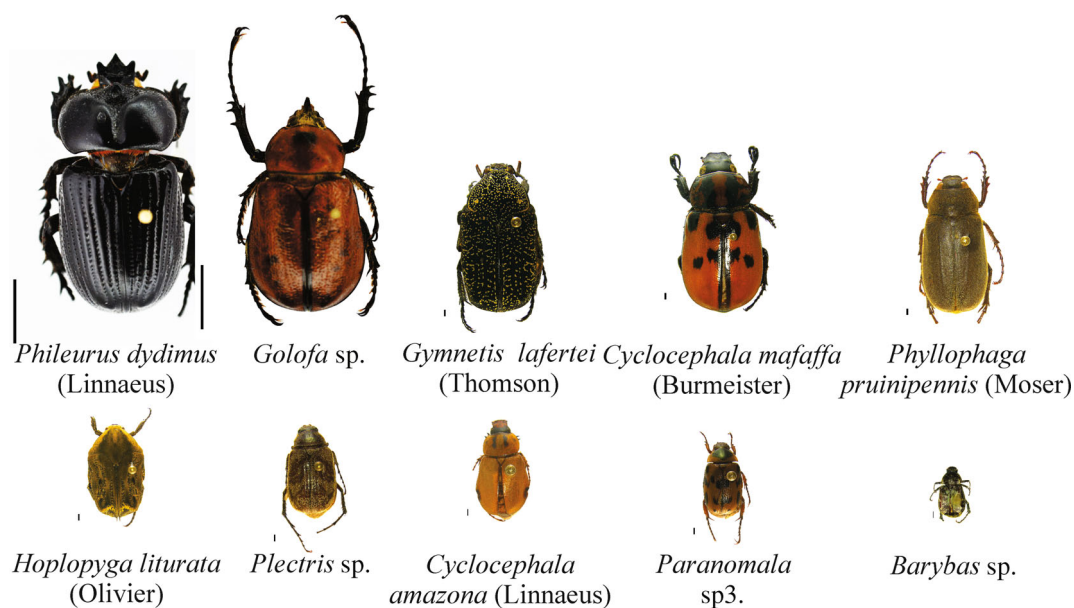


FIGURE 3 An example of the phytophagous scarab beetle species used to select functional traits. The species were sampled in the Serranía del Perijá, northern Colombia, and are scaled to show the variety of sizes and morphology.

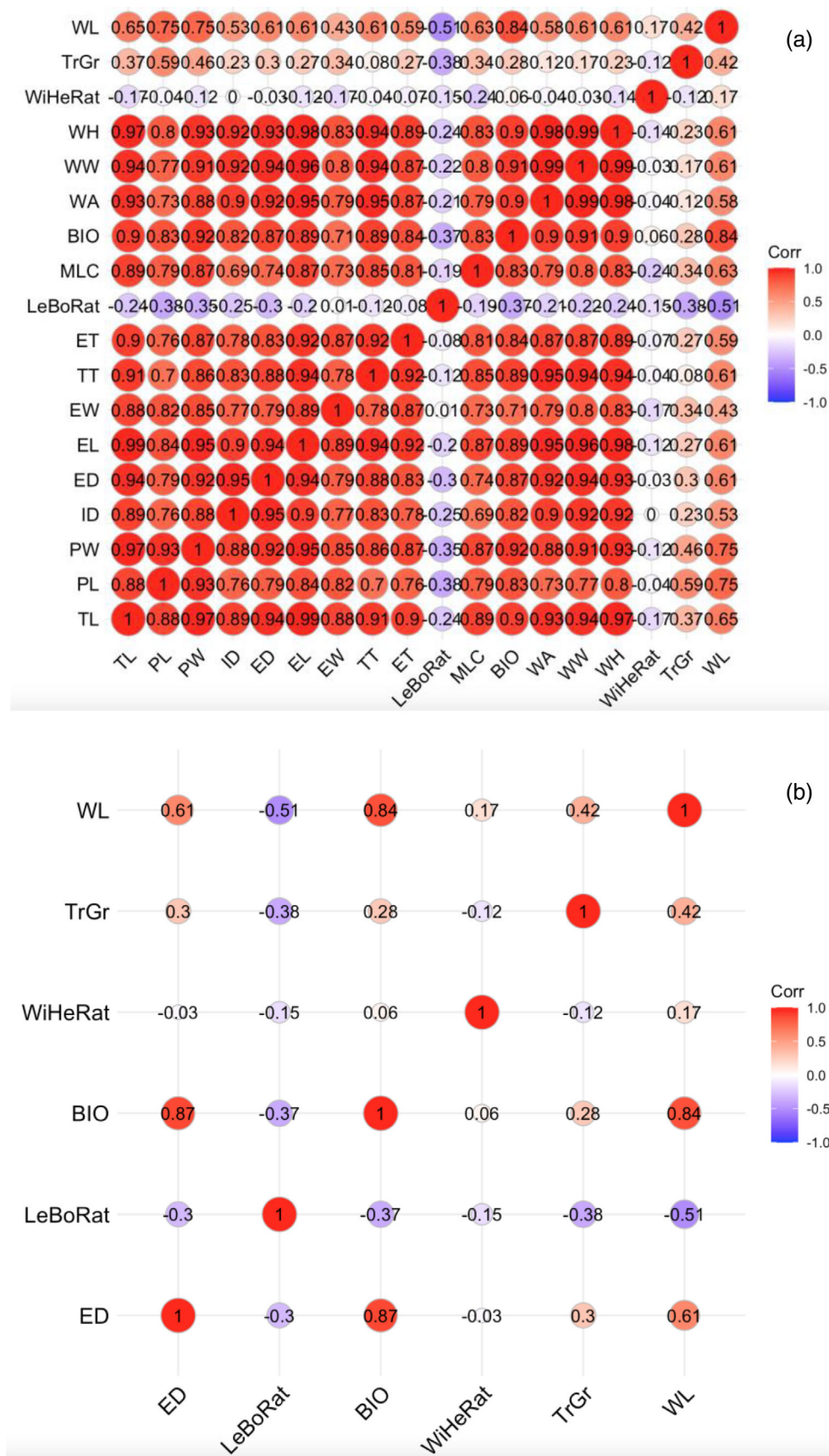


FIGURE 4 Correlation matrix of traits of the species in (a) broad proposal of traits and (b) traits chosen according to the correlation analyses. BIO, biomass; ED, eye diameter; EL, elytral length; ET, elytral thickness; EW, elytral width; ID, interocular distance; LeBoRat, leg length to body length ratio; MLC, maximum length of metacoxa; PL, pronotum length; PW, pronotum width; TL, total length; TrGr, trophic group; TT, total thickness; WA, wing area; WH, wing height; WiHeRat, wing width to height ratio; WL, wing loading; WW, wing width.

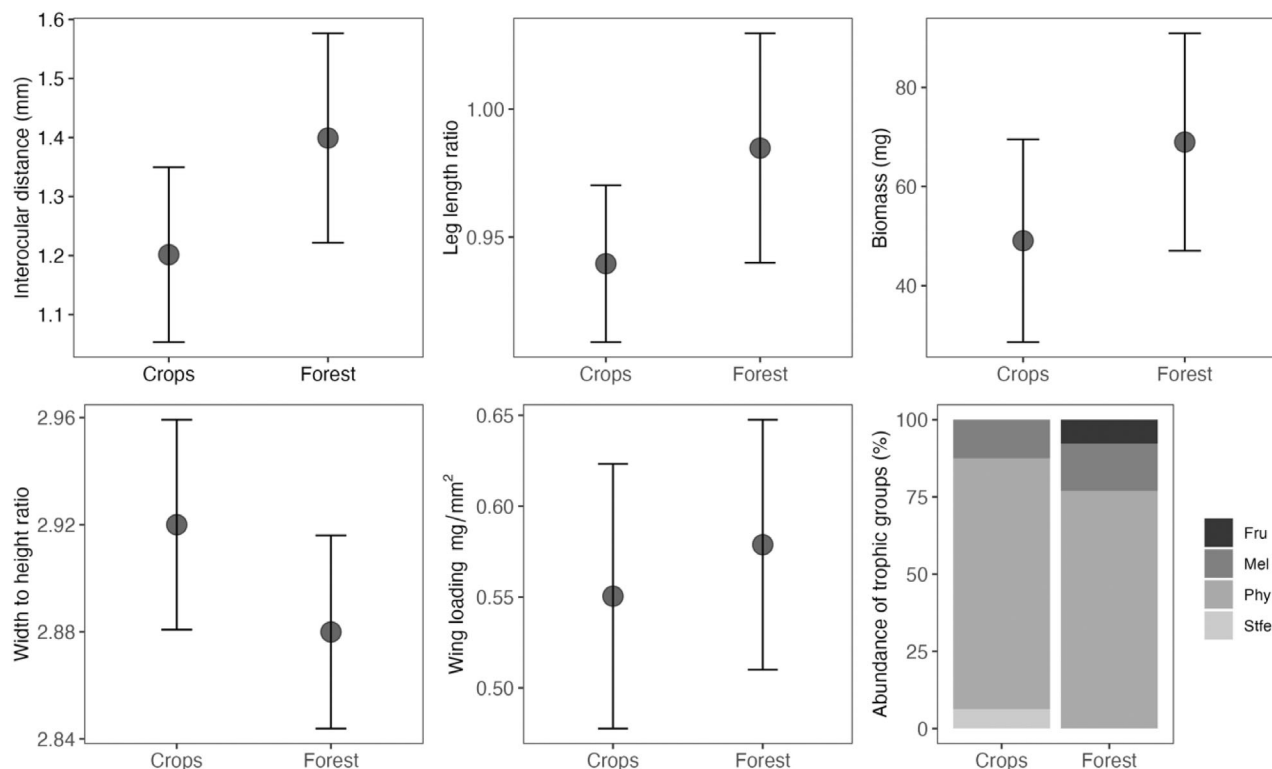


FIGURE 5 Variation of the six traits proposed as a minimal pool of traits for phytophagous scarab beetles, between coffee crops and forest sites in a landscape at the Serranía del Perijá, northern Colombia. The dot is the mean value and error bars are standard deviations. Trophic groups: Fru = frugivores, Phy = phytophagous, Mel = meliphagous, Stfe = stem-feeding species.

should be further evaluated in future studies aimed to test their performance as response or effect traits in different environments.

Third, selecting only a minimal set of meaningful traits is a clear advantage that can save time in future studies. A selected set of traits also eliminates the need to take a significant number of measurements, knowing in advance that those selected traits provide non-redundant information to a community. In our study case, obtaining values of 12 morphometric traits was a time-consuming task with microscopic equipment, and we discarded this information giving the high correlation among traits. Future studies should experimentally fill the gap of information relating these traits with ecosystem functions. For example, assessing the link between eye diameter and insect biological control, wing morphology and pollination or increased primary productivity, leg to body ratio and bioturbation or organic matter decomposition and biomass and trophic group with all the ecological functions described above.

Fourth, we suggest considering the feasibility of using other traits in beetle trait-based ecological studies to better elucidate functional contributions, such as the development of mandibles, which are more complex to measure. This trait can be categorised into attributes of 'low,' 'medium' or 'high' development based on the size and shape of the teeth and is related to how beetles exploit food resources. Mandibles may be reduced in those that consume soft, liquid or semi-liquid foods (flowers or sweet fruits), as is the case for cetonines (Suárez & Amat-García, 2007). In contrast, mandibles are well-

developed in groups that require decomposing resources and/or have predatory habits. In our study case, we were not able to include mandible measurements as we opted for easily measurable and quantifiable traits. Although we acknowledge the importance of mandibles in the ecology of phytophagous scarab beetles, their extraction and shape analysis require skills and more specialised techniques to generate shape categories to be considered as attributes in a functional analysis.

Moreover, including larval trophic groups would contribute to the understanding of how species contribute to bioturbation, but this information is widely unknown for some groups. This could make their assignment impossible or result in significant inaccuracies. One option is to incorporate this information at lower taxonomic levels and/or to include species for which we have a good understanding of their natural history. This trait was considered by Basset et al. (2023) in their study on the functional contribution of the Dynastinae subfamily in Panama, and they argued that the larval trophic group was one of the most challenging variables to determine because of the lack of knowledge regarding many species.

CONCLUSIONS

One of the main challenges in selecting functional traits for phytophagous species of Scarabaeidae is the limited knowledge of their natural

history and ecological functions. However, by analysing the available information on behavioural, morphological and feeding traits, it is possible to identify traits to recognise groups that explain the functional niche utilisation within a specific context.

Having a functional justification for the selected traits is crucial, as it provides a criterion, in addition to statistical measures, for organised trait exclusion in case they exhibit high collinearity. However, the decision to include or exclude traits should be considered beyond the statistical outcomes. For example, among the traits widely used in insects, femur and tibia lengths are often considered separately. Nevertheless, merging them and having a standardised measure based on body length is more coherent regarding the use of morphological structure in movement for dispersion and/or potential for colonising new spaces. In addition, making these decisions allows for summarising the pool of traits, which will likely make it easier to understand single- and multi-trait analyses of functional diversity.

Although it is important to incorporate sufficient traits based on the phenotypic plasticity of species to avoid functional redundancy, some traits should be avoided (de Bello et al., 2021). For example, the occurrence of a species in a specific habitat is more related to environmental characteristics (connectivity and habitat quality) than to behaviour. Similarly, the diurnal or nocturnal habits of beetles are not useful when individuals are collected in traps. This does not provide insights into the temporal niche partitioning of species in the ecosystem, leading to misinterpretations. Therefore, it is essential to have a clear concept of a functional trait. Accordingly, as our knowledge of trait variation within phytophagous scarab beetles grows within the framework of functional ecology, more precise selections can be made, optimising time and resources.

AUTHOR CONTRIBUTIONS

Claudia E. Moreno: Conceptualization; writing – review and editing; supervision. **Sandy García-Atencia:** Conceptualization; investigation; formal analysis; writing – original draft. **Maria Argenis Bonilla-Gómez:** Conceptualization; writing – review and editing; supervision.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The dataset generated and analysed during this study is available in the Supporting Information of this manuscript.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Table S1. Traits of phytophagous Scarabaeidae species selected from the entomological collection of the University of the Atlantic, UARC. BIO, biomass; ED, eye diameter; EL, elytral length; ET, elytral thickness; EW, elytral width; ID, interocular distance; LeBoRat, leg length to body length ratio; MCL, maximum metacoxa length; PL, pronotum length; PW, pronotum width; TL, total length; TrGr, trophic group; TT, total thickness; WA, wing area; WH, wing height; WL, wing loading; WW, wing width.

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